Forecasting How Busy a Venue Will Be, and Counting How Many People Are There

Jayden Bansal · Flock Social LLC, Pennsylvania · Research summary, October 2026

Full paper (40 pages) and technical version (60 pages): flockcorp.com/research

Research questions

1. Can a system that watches each venue's own recent live readings beat the popular-times chart at saying how busy one venue will be at one hour? **2.** Can a model small enough to run on a credit-card-sized computer count the people in a room from a 160 × 120 thermal camera, so that busyness can be checked against real people without recording anyone?

What I built

- **The forecast in Flock,** my app for planning nights out. A data pipeline collects hourly live busyness readings for about 1,750 venues in Philadelphia, the Lehigh Valley and Miami. The served number corrects each venue's typical-week curve with a trailing offset from its recent readings and uses the latest reading directly. A gradient-boosted model with 106 inputs (weather, events, holidays, neighbours, venue attributes) is trained and gated against production, and ships only if it wins.
- **Owl,** a 718,550-parameter neural network that counts people on a FLIR Lepton 3.5 thermal camera in 14 to 16 ms on one Raspberry Pi 5 core, trained on a physics-based generator of thermal scenes and six licensed real datasets. It runs inside **Flux**, a venue sensor I designed, where only numbers leave the device, never an image (Figure 1).

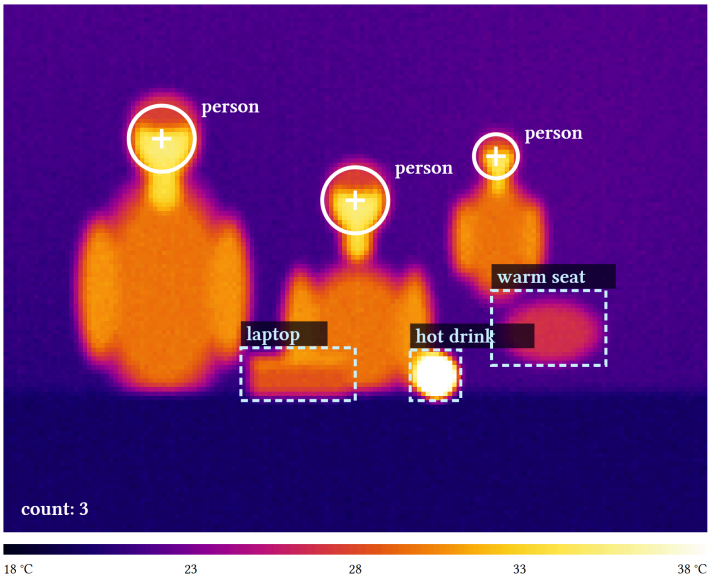


Figure 1. What Owl sees and what it reports. A 160 × 120 thermal frame in degrees Celsius, drawn the way Owl's training generator draws a scene, at the camera's true resolution. Owl puts a point on each person's head and a named box on every other warm thing, here a laptop, a hot drink and a seat someone just left, which a simple heat threshold would count as people. Only the points are counted: this frame counts as three.

How it was tested

Every forecast weight was fitted on earlier days and scored on **119,939 live readings** from 9 September to 8 October 2026 that it never saw, with date-block bootstrap intervals, persistence baselines and accuracy at every horizon from one to 24 hours. The forecast target is the provider's live reading, so the findings were then checked against real people: **5.7 million hourly counts from 573 counting sites in nine countries**. The core 5.3 million counts (315 sites: pedestrian counters, subway entries, a gym and a university library) were analysed under a plan whose pass rules were fixed and hashed before any data were downloaded; the rest add people counted inside rooms and libraries in Singapore, Barcelona and Karlsruhe. Owl was scored on six real benchmark sets and on sealed sets no version had seen.

Key findings

Result	Flock	Baseline
Exact crowd level (of five), 30-day test, one hour ahead	54.9%	30.2% popular times
Mean absolute error, same readings	18.00	26.40 (31.8% lower for Flock)
Exact level when the venue was read an hour earlier	77.0%	

Result	Flock	Baseline
Exact level two to 24 hours ahead	33.2–33.6%	30.2%
Real counts: one-hour error, curve + latest departure (15 datasets)	6.41	9.58 curve alone
People inside 153 campus spaces (Wi-Fi users): hour-old count wins at	152 of 153	typical-week curve
People counted in 5 rooms (ROBOD): one-hour error, hour-old count	12.18	21.63 curve
Owl, frames counted exactly, five of six benchmark sets	82–96%	9–77% heat rule
Owl, sealed set of 300 frames no version had seen	94.3%	40.7% heat rule

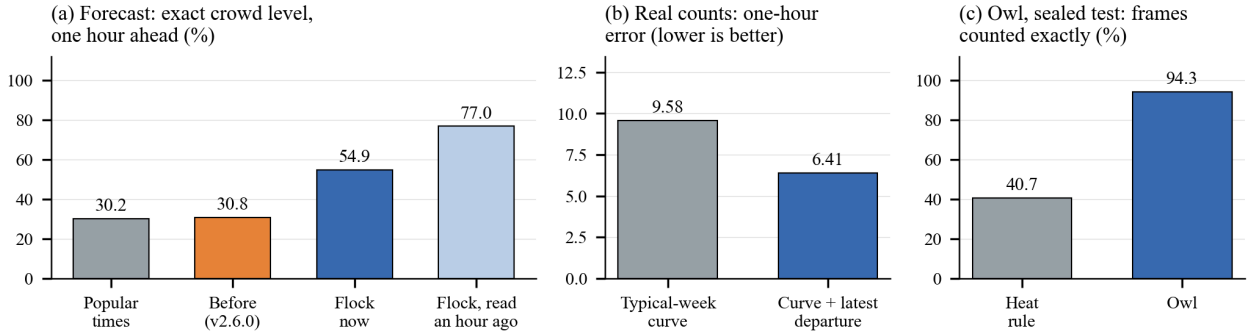


Figure 2. Headline results. (a) Share of one-hour-ahead forecasts that name the exact crowd level of five, on 119,939 held-out live readings. (b) Mean absolute error, one hour ahead, on the 15 pre-registered public real-count datasets (0 to 100 scale per site); carrying the latest departure from the curve onto the next hour was best in 14 of 15. (c) Owl against the heat-threshold rule on 300 sealed frames.

- **The freshest reading of a venue is worth more than any correction to its typical-week curve.** The served number's gain over popular times comes from using the latest live reading, and production now serves that reading directly.
- **The finding holds for real people.** Wherever the counts are people inside a place (library seats in Karlsruhe, 153 Wi-Fi-covered campus spaces in Barcelona, five rooms counted in Singapore), the hour-old count beats the typical-week curve, as it does on the provider's readings. On street and station counts the typical-week curve is close to calibrated (slope 0.92) and a departure from it lasts for hours, so carrying the latest departure onto the curve cut one-hour error by 33%. The provider's own curve tracks real movement too: beside 695 New York venues its weekly rhythm matches the nearest subway station's entries (median r 0.74).
- **Owl generalises across the cameras it was trained on and, in its stress tests, ignores the warm objects that fool a threshold** (hands, mugs, screens); on crowded classrooms it misses by one person on average, where its predecessor missed by almost five.

Scope

- Within Flock, the forecast is scored against the provider's relative 0 to 100 readings, not headcounts; the real-count datasets above are what tie its findings to people. The large gains are one hour ahead.
- Owl's first measurement inside a working venue, on Flux's own camera, is the next step; the benchmark sets that report its headline numbers also guided its development, and the sealed sets are the clean tests.

What comes next

A pre-registered retrain in early November 2026 on every reading collected, scored on 14 held-out days under a decision rule fixed before its data existed; a sealed test on Flux's own camera in real rooms of one to fifteen people; and a venue study that pairs hand counts with Owl and with the provider's live scores, to measure how commercial busyness scores track the people actually inside.

My role. I designed the research questions, built the data pipeline, the serving system and the sensor hardware, ran the experiments and wrote the software, using AI coding and writing assistants as tools throughout. Data: commercial busyness readings (provider not named), OpenWeather (ODbL), Ticketmaster, TheSportsDB; thermal datasets from TU Wien, IISER Bhopal, PUT, Aalborg University, ISSAI and the SenSys 2021 authors, used under their licences or with permission; public counts from 15 city, transit, library and university publishers, among them the KIT Library (with permission), credited in the paper.